

Infelicity with Homogeneity: Focus, not (necessarily) Presupposition*

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July 15, 2026

Abstract

A prominent but much-debated theory of the homogeneous behavior of sentences containing definite plurals is that homogeneity arises as the result of a presupposition. Wehbe (2022) and Guerrini and Wehbe (2024) present new arguments for this view based on the observation that definite plurals are infelicitous in certain contexts where one expects them to be infelicitous if homogeneity is a presupposition. We argue that closer scrutiny of the empirical picture reveals that infelicity in the cases they discuss is due to independent requirements from focus, and suggests the opposite conclusion, namely that sentences with definite plurals do not themselves give rise to a homogeneity presupposition. We further argue that, when such a presupposition does arise, this is due to the background question (QUD) rather than the sentence itself. We entertain a possible path to reconciling our arguments with the assumption that definite plurals trigger a homogeneity presupposition by making certain assumptions about presupposition accommodation.

Keywords: homogeneity · presupposition · focus · implicature · definite plurals · questions

1 Introduction

The phenomenon of homogeneity in definite plurals can be illustrated in (1). A basic question is how to explain the (near-)universal force of definite plurals in upward monotone environments as in (1a) and its (near-)existential force in downward monotone environments as in (1b) (Löbner 1987, 2000; Schwarzschild 1994; Krifka 1996; Gajewski 2005; Magri 2014; Križ 2015, 2016; Križ and Spector 2017; Bar-Lev 2021; Feinmann 2021; Chatain 2022, a.o.).

- (1) a. Ann read the books. *≈ Ann read **all** of the books*
 b. Ann didn't read the books. *≈ Ann read **none** of the books*

Some authors have claimed that homogeneity effects in definite plurals are the result of a homogeneity presupposition (HP), as in (2) (Schwarzschild 1994; Löbner 2000; Gajewski 2005, a.o.)

*For helpful discussion and comments we would like to thank Luka Crnič, Danny Fox, Roni Katzir, Benjamin Spector, Jad Wehbe, and the audience at the Gaps and imprecision in natural language semantics workshop in Berlin.

(2) **Homogeneity presupposition (HP):**

Ann read all of the books or none of them

(All $\vee$ None)

The HP theory is much debated, mainly surrounding questions of projection (Spector 2013; Križ 2015, 2016; Križ and Spector 2017; Fox 2018a). Homogeneity does appear to project like a presupposition in certain environments such as the scope of non-monotonic quantifiers. At the same time, it has been claimed that homogeneity projects differently from presuppositions in other environments, such as conditional antecedents. In addition, homogeneity doesn't seem to behave pragmatically like a presupposition, with respect to the *Hey, Wait a minute!* test (for discussion of the relevant facts see Spector 2013; Križ 2016; Križ and Spector 2017; Guerrini and Wehbe 2024). These differences have led to different perspectives on the status of homogeneity: as a plain presupposition (e.g. Schwarzschild 1994; Gajewski 2005), as a presuppositional implicature (Wehbe 2022; Guerrini and Wehbe 2024), or as a non-presuppositional inference (Križ and Spector 2017; Bar-Lev 2021).

Whether these observations provide conclusive arguments for or against the HP account remains an open question and we will have nothing to add about those particular issues here. What this paper focuses on is a new argument for HP that was recently put forth by Wehbe (2022); Guerrini and Wehbe (2024) (henceforth WGW) based on a property of presupposition accommodation. Specifically, WGW argue that assuming HP for definite plurals explains what we will call 'infelicity with homogeneity', illustrated by the infelicity of (3a) compared to the felicity of (3b), given that presupposition accommodation should not make the assertive component of meaning trivial; we will get into the details of their argument in the next section. (Throughout, capital letters represent focal stress. For now we will set aside the fact that stress on the quantifier is required in (3b) and other examples we discuss below; this will play an important role in our proposal and we will return to it in section 4.)

(3) I knew that Ann read at least some of the books. But today I learned that...

- a. #she read the books.
- b. she read ALL the books.

The goals of this paper are (i) to show that WGW's data can be explained without positing HP, and (ii) to show that closer scrutiny of the data in fact poses a challenge for HP. Our alternative account of (3a) relies on independently needed constraints on focus, and can explain variants of this infelicity where assuming HP would not help. We will however argue that in some instances of infelicity with homogeneity the infelicity is indeed due to HP, but that this HP is not triggered directly by the sentence but rather indirectly, from the (implicit or explicit) question that the sentence is taken to answer. Our focus-based alternative will then be compatible with various accounts of homogeneity when combined with standard assumptions about focus. But if our arguments against HP hold (and if other arguments in favor of HP that were mentioned above can be explained away), then they suggest that sentences with definite plurals should not directly trigger HP.

The note is structured as follows. In section 2, we review the argument by WGW for HP. In section 3 we argue that WGW's HP-based account of infelicity with homogeneity is both too weak (failing to predict infelicity in some cases) and too strong (predicting infelicity for some felicitous sentences). Section 4 then points out that there is a connection between infelicity with homogeneity and focus, motivating the alternative account we present in section 5 according to which infelicity with homogeneity is due to

conditions on focus. Section 6 is concerned with another kind of example that WGW brought up as an argument for HP; we argue that these are cases where HP arises from the background QUD rather than the sentence itself, and corroborate this by showing that the felicity of these examples is affected by focus structure. Section 7 discusses a possible direction for reconciling the HP view with our observations from section 3. Section 8 concludes and discusses infelicity with homogeneity in various domains where homogeneity effects have been observed.

2 The argument from PAI

Post-Accommodation Informativity (PAI) is a general principle that constrains sentences with a presupposition to only be felicitous if the assertive component of the sentence is informative after accommodating the presupposition (Heim 2015; Doron and Wehbe 2022).

(4) **Post-Accommodation Informativity (PAI):**

A sentence S_p (presupposing p) can be uttered felicitously only if S_p is informative w.r.t. the common ground and QUD after presupposition accommodation.

PAI explains why a case like (5a), involving the presupposition trigger *stop*, is infelicitous: as (6) shows, given the context-setting sentence and after the presupposition of (5a) is accommodated, there is no more room for the assertion encoded by (5a) to be informative.

(5) I knew that Mary doesn't smoke now. But today I learned that. . .

- a. #she stopped smoking.
- b. she used to smoke.

(6) In (5a): $\underbrace{\neg(\text{Mary smokes now})}_{\text{Context}} \wedge \underbrace{\text{Mary used to smoke}}_{\text{Presupposition}}$
 $\Rightarrow$ Mary used to smoke and doesn't smoke now

As previewed, Wehbe and Guerrini and Wehbe use PAI to furnish a new argument for a homogeneity presupposition for definite plurals. The paradigm in (7) is repeated from (3). The infelicity with homogeneity in (7a) is explained by PAI on the assumption that sentences with definite plurals trigger homogeneity as a presupposition. As (8) schematically illustrates, given the contextual setup which excludes worlds in which Ann read none of the books, (7a) becomes uninformative after accommodating the HP, explaining its oddness. By contrast, in (7b) there is no HP and consequently PAI isn't violated.

(7) I knew that Ann read at least some of the books. But today I learned that...

- a. #she read the books.
- b. she read ALL the books.

(8) In (7a): $\underbrace{\neg\text{None}}_{\text{Context}} \wedge \underbrace{(\text{All} \vee \text{None})}_{\text{HP}} \Rightarrow \text{All}$

A parallel reasoning applies to the paradigm in (9) in which the sentences are embedded under negation. HP+PAI for (9a) explains its infelicity, since the context-setting sentence contradicts **All**.

(9) I knew that Ann didn't read all of the books. But today I learned that...

- a. #she didn't read the books.
- b. she didn't read ANY of the books.

(10) In (9a): $\underbrace{\neg \mathbf{All}}_{\text{Context}} \wedge \underbrace{(\mathbf{All} \vee \mathbf{None})}_{\text{HP}} \Rightarrow \mathbf{None}$

3 Problems for HP+PAI

3.1 HP+PAI is too weak

A further welcome prediction of HP+PAI is that it correctly predicts sentences with definite plurals to be bad as answers to questions that arguably carry an existence presupposition:¹

- (11) a. Did Ann read some of the books, or all of them?
(Existence presupposition: Ann read some of the books)
- b. #Ann read the books.

This is because, as (12) shows, the response in (11b) is uninformative after accommodation of HP together with the existence presupposition introduced by the alternative question in (11a).

(12) In (11b): $\underbrace{\neg \mathbf{None}}_{\text{Existence presupposition}} \wedge \underbrace{(\mathbf{All} \vee \mathbf{None})}_{\text{HP}} \Rightarrow \mathbf{All}$

However, we point out that HP makes the wrong prediction for questions with no existence presupposition, where the facts look much the same:²

- (13) a. (i) Did Ann read none of the books, some of the books, or all of the books?
(ii) Did Ann read none of the books, or all of them?
- b. #Ann read the books.

HP+PAI doesn't explain (13), because the absence of the existence presupposition in the question does make the assertive component of the response informative, as schematized in (14).

(14) In (13b): $\neg \mathbf{None} \wedge \underbrace{(\mathbf{All} \vee \mathbf{None})}_{\text{HP}} \Rightarrow \mathbf{All}$

¹ We thank Danny Fox (p.c.) for pointing out to us the infelicity of sentences with definite plurals as answers to certain questions.

² As Jad Wehbe (p.c.) points out, one can consider a stronger version of PAI that rules out a sentence as an answer to a question if the sentence has a presupposition that eliminates cells in the question; this would explain why (13b) is bad as an answer to (13a-i) if it has HP. This version of PAI is supported by examples like the following (due to an anonymous reviewer):

- (i) Q: [I know that Ann owns either 2 or 3 cars.] Does Ann own 2 red cars, 2 blue cars, or 3 blue cars?
A: #Both of her cars are blue.

However, it would not explain why (13b) is bad as an answer to (13a-ii) as well, where the question already presupposes HP, and furthermore, it would not seem to help explaining the infelicity of (15). Crucially, when an answer presupposes what the question does, no infelicity arises:

- (ii) Q: [I know that Ann owns 2 cars.] Does Ann own 2 red cars, or 2 blue cars?
A: Both of her cars are blue.

Another example that shows that HP+PAI is too weak is the following (due to Luka Crnič, p.c.):

- (15) #Yesterday she read some of the newspapers. Today she read the newspapers.

Note that the first sentence in (15) does not entail anything about what happened today, and specifically it does not entail that she read some of the newspapers today; but the second sentence is still infelicitous. There is thus reason to cast doubt on the generality of WGW's HP+PAI account of infelicity with homogeneity in section 2.

3.2 HP+PAI is too strong

Another problem which looks even more concerning for HP+PAI is that HP+PAI over-predicts infelicity for certain felicitous utterances. Consider (16), involving polarity ('verum') focus.^{3,4}

- (16) A: I know that Mary read some of the books, but I suspect that she didn't read all of them.
B: No, don't worry, she DID read the books.

The felicity of (16) is unexpected if we assume HP+PAI, because given A's context-setting sentence, B's assertion should be uninformative. Importantly, run-of-the-mill cases of PAI violations with uncontested presuppositions do not behave like this. (17)B illustrates with a PAI violation stemming from the presupposition of *stop*:⁵

- (17) A: I know that Mary doesn't smoke now, but I'm worried that she also didn't smoke in the past (in which case she is not a good fit for our experiment).
B: #No, don't worry, she DID stop smoking.
B': No, don't worry, she DID smoke in the past.

Given that PAI seems to hold quite generally, the felicity of (16)B then poses a problem for the assumption that sentences containing definite plurals trigger HP. In section 5 we will discuss another example that poses a similar problem to HP (ex. (39)).

3 Adding a numeral in the definite plural (*the five books* rather than *the books*) does not affect the judgment for (16), and will guard against a possible explanation of its felicity based on an analysis of *the books* that involves domain restriction to the books that the speaker doesn't know whether Mary read.

4 (16) sounds entirely natural to us and to several speakers we consulted with; however, a reviewer judges this sentence to be bad, but also points out that their judgment could be because the phonologically reduced *read the books* in B's utterance does not have matching material in A's utterance. As they point out, this can be avoided by having the following discourse, which makes the same point as (16):

- (i) A: I suspect that Mary didn't read the books. I mean, she did read some of them, but. . .
B: No, don't worry, she DID read the books.

5 Focus is sometimes able to rescue from PAI violations, as the felicity of (i) in the context of (17) demonstrates. We will get back to this point and its relevance to the HP theory in section 7.

- (i) B'': No, don't worry, she STOPPED smoking.

3.3 Intermediate summary

To take intermediate stock, the assumption that sentences with definite plurals trigger HP can explain certain cases of infelicity with homogeneity, but it appears to be both too-weak (section 3.1) and too-strong (section 3.2). In the next two sections we will propose an alternative explanation of GWG's facts. We will point out that independent conditions on focus placement seem to underlie the behavior of the felicitous and the infelicitous examples we've seen so far. This will lead us to present a focus-based analysis of the facts which does not assume HP.

4 Towards an alternative account: the homogeneity-focus generalization

As a first step towards an analysis, we want to point out that there is a close connection between infelicity with homogeneity in the examples we discussed so far and the stress pattern observed when the intended meanings are conveyed by overt quantifiers. As it turns out, in all the cases we have seen, when there is an overt quantifier, it must be stressed for the sentence to be felicitous. That is, the following generalization seems to hold:

(18) **The homogeneity-focus generalization:**

A sentence with a definite plural in a context *C* is infelicitous if its overt-quantificational counterpart requires stress on the quantifier in order to be felicitous in *C*.

Let us show that the generalization in (18) captures all the cases of infelicity with definite plurals that we have seen so far. The infelicity in (19a), (20a), (21b-i) and (22a) correlates with the fact that the quantifier in (19b), (20b), (21b-ii) and (22b) must be stressed for the sentence to be felicitous (observe that for the definite plural version of the examples, by contrast, there is no stress pattern that makes them felicitous; in particular, stress on the definite article doesn't save the sentences).

(19) I knew that Ann read at least one of the books. But today I learned that...

- a. #she read the books. [= (3a)/(7a)]
- b. she read {ALL, #all} the books.

(20) I knew that Ann didn't read at least one of the books. But today I learned that...

- a. #she didn't read the books. [= (9a)]
- b. she didn't read {ANY, #any} of the books.

(21) a. (i) Did Ann read some of the books, or all of them?
(ii) Did Ann read none of the books, some of the books, or all of them?
(iii) Did Ann read none of the books, or all of them?

- b. (i) #Ann read the books. [= (11b)/(13b)]
(ii) Ann read {ALL, #all} the books.

(22) a. #Yesterday she read some of the newspapers. Today she read the newspapers. [= (15)]

- b. Yesterday she read some of the newspapers. Today she read {ALL, #all} the newspapers.

The homogeneity-focus generalization also implies that if a sentence with a definite plural is felicitous, then its overt-quantificational counterpart does not require stress on the quantifier. This is indeed borne

out when we consider (16), as (23) demonstrates:

- (23) A: I know that Mary read some of the books, but I suspect that she didn't read all of them.
 B: No, don't worry, she DID read the books. [= (16)]
 B': No, don't worry, she DID read all the books.

5 Focus as the culprit

5.1 Some common assumptions on focus

The homogeneity-focus generalization suggests that the infelicity with homogeneity in all the cases we discussed has to do with considerations of focus. As a clue to why that should be, we may first ask why overt quantifiers require stress in (19)–(23). The answer looks rather obvious and relates to the known connection between focus and discourse, in particular (under one implementation) between focus and Questions Under Discussion (QUD; Roberts 1996a): The contexts in all of these examples make it so that the most natural QUD cares about the distinction between quantifiers (*none/some/all*); we will call this type of question a ‘quantificational question’. This is immediately visible when such a question is made explicit (as in (21)); in this case standard assumptions about focus require focus on the quantifier, as in (24b).

- (24) a. Did Ann read none of the books, some of the books, or all of them?
 b. Ann read [all]_F of the books.

Specifically, the requirements on focus placement that we adopt are in (25). They make it so that (24b) is the only possible F-structure of the sentence: *all* must be dominated by an F-marked constituent given a *Q* like (24a) in order to satisfy the congruence requirement in (25a) (see Rooth 1992, Roberts 1996b, Büring 2019, Katzir 2024, a.o.), and F-marking on any constituent other than *all* would violate the requirement on focus minimization in (25b) (Schwarzschild 1999):

- (25) **Focus Placement:**
- a. **Question-answer congruence:** The focus structure of a sentence *S* must be congruent to an explicitly-provided or implicitly-accommodated question *Q* that *S* answers, such that $Q \subseteq \llbracket S \rrbracket_f$.
 - b. **Focus minimization:** F-mark as little as is necessary to satisfy question-answer congruence.

We assume that *some* and *none* are focus alternatives of *all*.⁶ The focus value of (24b) is then as in (26), which is congruent to the question in (24a).

- (26) $\llbracket (24b) \rrbracket_f = \{ \text{Ann read none of the books, Ann read some of the books,} \\ \text{Ann read all of the books} \}$

⁶ There are reasons to assume that *none* is not a formal focus alternative of *all*, given arguments that it is composed of negation and an existential quantifier (Zeijlstra 2011; Penka 2011, a.o.) and under the assumption that focus alternatives are no more complex than the prejacent (Katzir 2007; Fox and Katzir 2011). But even on these accounts it should be an alternative given the question in (24).

Furthermore, given the following constraint on the realization of focus, *all* would have to be stressed given the structure in (24b) (see Truckenbrodt 1995; Büring 2016, a.o.):

(27) **Focus Realization:**

The highest stress in a sentence (more generally, in a focus domain) lies on an F-marked phrase in the sentence. (adapted from Büring 2016:164)

5.2 Explaining infelicity with homogeneity in answers to quantificational questions

We will start by analyzing cases in which the target sentences are used as answers to explicit questions. This will make it easier to appreciate the core aspects of our theory. We will turn in section 5.4 to cases with no explicit question. The first order of business is then to explain why we get infelicity with homogeneity in answers to quantificational questions like the following:

- (28) a. Did Ann read none of the books, some of the books, or all of them?
b. #Ann read the books.

A natural explanation for the infelicity of (28b) suggests itself: the focus requirements cannot be met due to the lack of an overt quantifier. Our proposal is based on this intuition: in a nutshell, the lack of an overt quantifier against a quantificational QUD means that either F-marking does not satisfy the F-placement requirements in (25) to begin with, or, even if it does, it will fail to be realized phonologically, violating the constraint on F-realization in (27).

This line of explanation is consistent with all theories of how homogeneity actually arises. In all of them, there is nothing overt in the structure of (28b) that contributes quantification. On some accounts the quantificational force of the sentence comes from extra-syntactic sources (e.g., Križ 2015; Križ and Spector 2017), which means that there is no way to F-mark anything in the structure so that F-placement would be satisfied if *Q* is a quantificational question; in particular, no matter how focus is assigned, it will be impossible to derive focus alternatives that are equivalent to the *some*, *none*, and *all* alternatives in the question.⁷ On other accounts there is a covert operator in the structure that introduces quantification (e.g., Gajewski 2005; Bar-Lev 2021; Guerrini and Wehbe 2024), which under certain assumptions about its focus alternatives might make it so that F-placement can be satisfied, but even in that case F-realization won't be satisfiable due to the covert nature of the operator. In the rest of the paper we will assume that the infelicity is due to failure to satisfy F-realization. As a reviewer points out, if failure to satisfy F-realization is the source of infelicity, we have to assume that it is an inviolable constraint; this is not a trivial assumption given that F-realization is an interface constraint, but it is one we share with Büring (2016: p. 184). See appendix A for elaboration on the choice between blaming F-placement and blaming F-realization within theories that assume a covert operator and some of the theoretical choices one can make in this regard.

⁷ Though, for such accounts to ensure failure of Congruence in the relevant cases, they would have to commit to the assumption that focus alternatives are constrained by syntactic complexity, as in Fox and Katzir (2011). If those accounts assumed instead that focus alternatives are constrained by semantic type, as in Rooth (1992), then one could always find a way to make the answer congruent, e.g., by having focus on the whole VP: If focus alternatives are anything of the same semantic type, the set of focus alternatives of *Ann [read the books]_F* would contain the quantificational alternatives (because *read the books* and *read none/some/all of the books* are of the same semantic type). And since focus minimization would not force any smaller focus in the absence of a covert quantificational operator that triggers these alternatives in the sentence, this would be a licit choice of focus structure, which could easily satisfy F-realization. We thank a reviewer for pointing this out to us.

5.3 Issues of Question Accommodation

Before we continue to show how what we said about (28b) can explain infelicity with homogeneity more generally, we would like to discuss a potential objection that may arise for the idea that what's at stake here is a violation of either F-placement or F-realization. There is a simple-minded explanation that one may suggest for the infelicity of (28b) that has to do with neither of these constraints, and instead has to do with the requirement that the answer should be syntactically identical to a member of the question. We however note that this cannot be the whole story; sentences with definite plurals can in fact felicitously answer questions with quantifiers in general, as long as the question is not a quantificational one. Consider the following question-answer pairs:

- (29) a. Q: Did she read all the books?
 A: (Yes,) she (DID) read the books
 b. Q: Did she not read all the books?
 A: (No,) she DID read the books

The answers in (29) are perhaps not perfect, and one may prefer variants of these questions with *all*, but they are much better than (28b). It seems that, even if there is indeed a syntactic identity requirement, it can be obviated when the question is non-quantificational, as in (29), but cannot be obviated when the question is quantificational, as in (28b); this is explained if the failure in (28b) has to do with constraints on focus, and not (only) with syntactic identity, as we assume.

A possible way to think about the felicity of the answers in (29) is that, even if there is a syntactic identity requirement, it can be obviated by accommodating a different question than the explicit one, one for which the answer satisfies the syntactic identity requirement. This raises the general question why it is impossible to rescue the answer in (28b) by accommodating a different question than the one asked explicitly; this question doesn't only arise with respect to syntactic identity, but also with respect to the constraints on focus that we assumed above: even if these constraints are inviolable, one would still expect them to be satisfiable if speakers could accommodate a different question, one which would satisfy these constraints. This looks possible to do in various cases. For instance, consider the following:

- (30) Q: Do you want coffee, or tea?
 A: I want water.

Presumably, here the answer is felicitous because A accommodates a question like *what do you want?*, with *what* ranging over coffee, tea, and water; this question is related to the explicit question but is also crucially different from it. The challenge is then to explain why the indirect answer in (30) sounds felicitous while the one in (28) doesn't. The infelicity of (28) compared to the felicity of (30) is in fact all the more surprising, given that (28b) is as close as one can get to a direct answer to the explicitly posed question: it is equivalent or near-equivalent to its respective universal-quantificational counterpart, which is one of the mentioned alternatives in the question. The same cannot be true for the discourse in (30). So if anything, one would expect the inverse pattern of felicity than the one actually observed.

Let us then ponder what types of questions one could in principle accommodate in order to save (28b). Certain types of questions, such as those in (31), won't be of any help here:

- (31) a. How many books did Ann read (if any)?

- b. Which book(s) did Ann read (if any)?

This is because a non-quantificational answer to these questions is infelicitous as well, just like in (28b), and a quantificational answer to these questions requires stress on the quantifier, as in (24b) (in line with the homogeneity-focus generalization). Apparently, then, accommodating these questions as the background question for (28b) would lead to a violation of the constraint on focus, just like the original question in (28a) (in fact, one may suspect that answering these questions with *Ann read all the books* requires accommodating a quantificational question, at least in the case of the singular *which* question, which is presumably incongruent with this sentence).

What about other questions? Obviously, sentences with definite plurals are often good answers to other questions, and the stress pattern they require depends on the question (as we already observed in (29)). For instance, if the question is (32a) or (32b), the appropriate stress pattern for *Ann read the books* would be default stress (or polarity stress, in the case of (32b)); if it is (32c), then it would be stress on *Ann*, and if the question is (32d), the appropriate stress for an answer like *Ann read the blue books* would be stress on *blue*.⁸

- (32) a. What happened?
 b. Did Ann read the books (or not)?
 c. Who read the books?
 d. Did Ann read the blue books, or the red books?

What distinguishes these questions from (28) is that none of them is a quantificational question, and answering them with *Ann read the (blue) books* is perfectly fine. We then have to assume that these questions cannot be accommodated in the face of explicit quantificational questions (but see footnote 9).

Providing a full account of when question accommodation is licensed is beyond the scope of this paper, but we would like to emphasize that, for our purposes, it suffices that the unavailability of accommodating the questions in (32) in the face of explicit quantificational questions is not specific to the non-quantificational sentences that are the focus of our discussion, i.e., it's not specific to homogeneity; an entirely parallel assumption is needed also in order to explain why the quantificational variants require stress on the quantifier, i.e., why (24b) is infelicitous unless stress falls on *all*, something which is

⁸ A reviewer makes the intriguing observation that, while the quantificational question in (13a-ii) (=Did Ann read none of the books, or all of them?) is expected to be semantically identical to the polar question in (32b) given homogeneity, the stress pattern of their answers is still different:

- (i) a. As response to the question in (13a-ii): Ann read #(ALL) the books
 b. As response to the question in (32b): Ann read (#ALL) the books

This is not expected if congruence only cares about the semantics of the question and the members of the focus value of the sentence, rather than the form. We would however like to point out that this effect is not specific to homogeneity. Given the context below, the questions in (iia) and (iib) are identical, yet the stress pattern of their answers is different.

- (ii) Context: We know that only one person is in love with Ann, but we don't know whether it's Sue or Mary.
 a. Q: Is Sue in love with Ann, or is Mary in love with her?
 A: {SUE, #Sue} is in love with her.
 b. Q: Is Sue in love with Ann (or not)?
 A: {#SUE, Sue} is in love with her.

We remark that both (i) and (ii) can be explained if Congruence cares about the syntactic form of the question rather than (only) its semantics in context, and if the target question in each case is the one explicitly asked in its respective context.

not expected if one could accommodate just any implicit question (e.g., *Did Ann read all the books?*). Presumably, any theory of why this stress pattern is obligatory with overt quantificational answers and cannot be avoided by question accommodation would also explain why the non-quantificational answers are impossible to rescue with question accommodation.⁹ In what follows we will continue to take the behavior of overt quantifiers as our guide: While we cannot directly observe what questions one can accommodate, we can indirectly infer this from the stress pattern of sentences with overt quantification: Whenever we see that the quantifier must be stressed, we infer that a non-quantificational question cannot be accommodated.

5.4 Explaining infelicity with homogeneity in the absence of explicit questions

In the infelicitous examples discussed by WGW, such as (19a), there is no explicit question; our focus-based explanation for the infelicity in (28) then does not immediately explain the infelicity in the absence of explicit questions. We can however extend this analysis to WGW's examples if we can show that in the relevant contexts one cannot (naturally) accommodate a question that isn't a quantificational one. How can we show this? As above, we take the stress pattern in the overt-quantificational answers as our guide: Whenever stress must fall on the quantifier, this indicates that the background question must be quantificational, and in that case the infelicity of the non-quantificational answer would be reduced to the infelicity in (28). And conversely: Whenever stress does not have to narrowly fall on the quantifier in the overt-quantificational counterpart, this indicates that the question is not a quantificational one, in which case we should expect the non-quantificational variant to be acceptable (assuming it does indeed answer the background question well; in section 6 we will discuss examples where this is not the case).

We hold that this approach allows us to extend our account of (28) above to all the infelicitous cases with definite plurals we've seen so far, with or without an overt question, cf. our discussion in section 4 (examples (19a), (20a), (21b-i) and (22a)). In all of them, the discourse is structured in a way that leads us to accommodate the offending quantificational question, even where no such question is overtly provided, as corroborated by the obligatoriness of stress on the quantifiers in the overt quantificational counterparts. Take WGW's original example, repeated in (33):

⁹ As a clue as to why that could be the case after all, we point out that certain discourse moves that would be required for question accommodation in the relevant examples are strange, as the contrast between the following examples demonstrate:

- (i) a. Did she read some of the books, or all the books?
b. #I'm going to answer the (slightly different) question of {what happened, whether she read the books}. She read (all) the books.
- (ii) a. Which book(s) did Ann read?
b. I'm going to answer the (slightly different) question of whether she read some of the books or all of the books. She read ALL of the books.
- (iii) a. Do you want coffee, or tea?
b. I'm going to answer the (slightly different) question of whether I want coffee, tea or water. I want WATER.

We also note that it is not absolutely impossible to imagine special scenarios where the discourse in (28) will sound fine, but such scenarios presumably involve an interlocutor who is generally oblivious to norms of conversation. Specifically if we imagine the speaker of (28b) to be a person who dismisses or completely ignores the question raised by the speaker of (28a), then (28b) may ring felicitous, with the understanding that the speaker then answers one of the accommodated questions in (32) (depending on the specific stress pattern).

- (33) I knew that Ann read at least one of the books. But today I learned that...
- a. #she read the books. [= (3a)/(7a)/(19a)]
- b. she read {ALL, #all} the books.

The infelicity of (33a) stems from the fact that the preceding discourse in (33) leads one to most naturally accommodate a quantificational QUD: {*Ann read some of the books, Ann read all of the books*}. This question supports no QUD for which F-placement and F-realization can both be met in (33a). We conclude, then, that considerations of focus explain infelicity with homogeneity in all examples we have seen so far in a way that is not dependent on any specific theory of homogeneity. This undermines WGW's argument in favor of HP, though it is in principle compatible with the existence of HP.

5.5 Explaining felicity with homogeneity in the absence of explicit questions

Recall however that (16), repeated in (34), is felicitous despite HP+PAI predicting that it should be infelicitous, thus providing an argument against HP.

- (34) A: I know that Mary read some of the books, but I suspect that she didn't read all of them.
B: No, don't worry, she DID read the books.

Our focus-based proposal, in contrast, accounts for the felicity in (34) by considering the type of background question that is licensed by the context and the focus structure. Let us see how. We have already seen in section 4 that, in line with the homogeneity-focus generalization, in the context in (34) the parallel sentence with an overt quantifier does not require stress on the quantifier (ex. (23)):

- (35) B': No, don't worry, she DID read all the books.

Given what we just said, this means that the context must support a non-quantificational question. A natural candidate for such a question is the polar (non-quantificational) question in (36); this question is also what one would expect the focus value of (35) to be, given what's usually assumed for polarity focus (e.g. Laka 1990; Wilder 2013; Roelofsen and Farkas 2015; Goodhue 2022).

- (36) {*She read all the books, She didn't read all the books*}

We propose that also for the utterance in (34), the context supports the same background polar question. We thus expect felicity here for the same reasons that the following question-answer pair, repeated from (29b), is felicitous:^{10, 11}

¹⁰ The obligatoriness of polarity focus here is expected given the general observation that answers to polar questions with the opposite polarity than the question require polarity focus (Goodhue 2022). This extends to discourses that are similar to (34), where there is no explicitly asked question but the naturally accommodated question has negative polarity:

- (i) A: I know that Mary read some book but I suspect she didn't read War and Peace (as she was required to).
B: No, don't worry, she {#read, DID read} War and Peace.

¹¹ One may wonder how the second member of (36) ends up in the focus value of *she DID read the books* in (34)B, that is, how Congruence as defined in (25) is satisfied here in the absence of overt *all*. Here are two ways to achieve this (the choice between these two options does not matter here but will have repercussions down the line; see fn. 18). First, if one adopts the assumptions in Katzir (2007); Fox and Katzir (2011) concerning focus alternatives, *she didn't read all the books* is a focus alternative of (34)B by being mentioned in the discourse, satisfying the condition $(36) \subseteq \llbracket (34)B \rrbracket_f$. Second, on the implicature account of homogeneity (Magri 2014; Bar-Lev 2021), one would be able to derive an alternative equivalent to the *not all* sentence by

- (37) Q: Did she not read all the books?
 A: (No,) she DID read (all) the books

On our proposal, the felicity of (34), as opposed to the previous cases with definite plurals, stems from the fact that the salient QUD in this context plus the rules of focus realization conspire to allow it. Specifically, the polar QUD calls for polarity focus, and polarity focus is independently realized (in English) with stress on the auxiliary. This configuration then allows the definite plural to exhibit its homogeneous behavior, i.e. its denotational equivalence with *all* in UE environments, without causing a crash with the grammar of focus: Congruence is satisfied for (34) against the QUD in (36) just as much as it is satisfied for (35), and stress predictably falls on the auxiliary (in both (34) and (35)).

A reasonable question to ask at this point is why a polar question like (36) is not available in the contexts discussed by WGW, such as (33) (note that polarity stress does not seem to help in these examples). We do not know why that happens; but we note once again that, whatever might be the reason for that, the unavailability of such polar questions in their contexts is supported by the obligatory stress on *all* in the overt quantificational variants in their examples.

The point we're making with example (34) is in fact general and does not affect just polarity focus. Whenever the background question and focus calculus independently predict that main stress should fall on some other overt element in the sentence, our proposal predicts that a definite plural will be fine in a PAI-like context, whereas WGW's HP would still predict infelicity. Let us now demonstrate that our account makes the right prediction for cases other than polarity focus, using a slightly more involved kind of example and with a richer context. As a baseline, consider first (38a), which is odd no matter the prosody when uttered in the context of (38).

- (38) John really likes to write phonology papers. Every day, he either works on some of them or all of them. He is in a cheerful mood now, so apparently. . .
- a. #he worked on his phonology papers today.
 - b. he worked on {ALL, #all} of his phonology papers today.

(38a) is essentially a replication of WGW's example in (33a). Just like in (33a), HP+PAI predicts infelicity, since the context entails that he worked on some of his phonology papers today; and just like in (33a), the most natural question in (38) is a quantificational question, as corroborated by the stress pattern in (39b), so one can also explain the infelicity of (39a) based on considerations of focus. Now consider the more elaborate context in (39), where the same sentence looks fine if stress falls on *phonology*, as in (39a).

- (39) John really likes to write phonology papers. He also writes semantics papers, but he hates that. Every day, he either works on all of his phonology papers and none of his semantics papers, or all of his semantics papers and some of his phonology papers. He is in a cheerful mood, so apparently. . .
- a. he worked on his PHONOLOGY papers today.

letting the exhaustivity operator that derives the universal interpretation for sentences with definite plurals take scope below the polarity head and replace the positive polarity with a negative one. On a view where there is HP, making (36) congruent with (34)B in a similar way would require locally accommodating the homogeneity presupposition of (34)B below the polarity head. We will discuss the implications of having local accommodation in section 7.

- b. he worked on all of his PHONOLOGY papers today.

For HP+PAI, there should be no difference between (38) and (39): In both of them the context entails that John worked on some of his phonology papers today, leading to a violation of PAI. The fact that (39) is felicitous is then surprising for HP+PAI, and, like (34), suggests that HP is not semantically encoded.¹² On our focus-based alternative, on the other hand, the felicity of (39a) is explained assuming that the most natural QUD in this context is as follows, as indicated by the stress pattern in (39b):

- (40) {*He worked on all of his phonology papers today*,
He worked on all of his semantics papers today}

Given this, the structure of the explanation carries over from the earlier case with polarity focus. (39a) and (39b) are denotationally equivalent given homogeneity; and (39a) is just as good as (39b) because, as opposed to their occurrence in the context in (38), the contextually-appropriate QUD is the one in (40), and the focus realization rules make it so that all conditions on F-marking are met, as now F-marking legitimately falls on overt material (*'phonology'*).¹³

To sum up, all examples of infelicity with homogeneity so far are explained given some common assumptions about focus; these assumptions explain not only the infelicity of some examples that are explained by HP+PAI ((19a) and (20a)), but also the infelicity of examples that are not explained by HP+PAI ((21b-i) and (22a)) and the felicity of some examples that HP+PAI wrongly predicts to be infelicitous ((34) and (39a)). As we pointed out, the focus-based explanation does not require theoretical commitment to any particular theory of homogeneity, though if one would like to enjoy its benefits in accounting for the felicity of (34) and (39a), one would presumably want to stay away from accounts in which HP is triggered by sentences with definite plurals (but see section 7).

6 Question-induced HP and infelicity

On our view, infelicity with definite plurals stems from inability to satisfy conditions on focus given the explicit or reconstructed QUD. In the infelicitous cases where a question had to be reconstructed, we argued that it is the stress pattern of the quantificational version of our definite plural sentences that served as clue as to what questions can be reconstructed: a quantificational question can be inferred to be the only question that can be accommodated when stress obligatorily falls on the quantifier, and given that only such a question can be accommodated we could explain why the definite plural version was infelicitous by appealing to conditions on focus. Challenges to our view would therefore come from definite plural examples that are infelicitous even though the stress pattern of the quantificational

¹² One may suspect that the inference from (39a) that he worked on none of his semantics papers today is enough to make the sentence non-trivial and thus avoid violating PAI. Note however that the context is set up in such a way that it is guaranteed that if he worked on all of his phonology papers than he worked on none of his semantics papers. So once HP (**All phonology** $\vee$ **No phonology**) is combined with the contextual assumptions (**Some phonology** $\wedge$ (**All phonology** $\rightarrow$ **No semantics**)), it follows that **All phonology** $\wedge$ **No semantics**, leaving no room for the sentence to be informative.

¹³ For some speakers the judgment regarding the felicity of (39a) might be subtle. We submit nonetheless that there is a contrast between the occurrence of this sentence in (39) vs. its occurrence in (38). The subtlety may be attributed to the fact that the context in (39) is intricate and that accommodating the right QUD from the sentence is somewhat of a complicated task. A reviewer pointed out to us that another potential reason why it may not be entirely natural is that some speakers may have a reading of *work on his papers* that is equivalent to *work on some of his papers*, in which case (39a) would be contextually redundant.

version does not appear to trigger reconstruction of a quantificational question. Let us now discuss two challenging examples from this angle. One example is (41), which WGW bring up (due to Roger Schwarzschild, p.c.) as a further argument for HP+PAI.

- (41) *Context: A monogamous society.*
- a. #John is not married to his teammates.
 - b. John is not married to any of his teammates.

This minimal pair involves negation, and definite plurals under negation are (normally) interpreted existentially. Yet, there is a stark contrast between (41a) and (41b). HP+PAI correctly predicts the infelicity of (41a), since the context entails that John is not married to all his teammates (a monogamous society), and this means that the sentence violates PAI: the conjunction of $\neg$ **All** (context) together with the **All** $\vee$ **None** presupposition (HP) makes the sentence (**None**) trivial. This is the same structure of explanation that was given to an example we saw earlier, (9a); there too a negative sentence with a definite plural is infelicitous.

From the point of view of our analysis, the distinctive feature of (41) is that, differently from (9), the context doesn't seem to require focus on the quantifier *any* in (41b) (cf. (9b)); we have default stress here, of the kind expected from an utterance in all-new contexts—and indeed, (41b) can be uttered more or less out of the blue. This means we cannot, in contrast to our analysis of previous cases, rely on the stress pattern of the quantificational version to determine that the background question is a quantificational one. Therefore we cannot plausibly explain the infelicity along the same lines as we explain (9), i.e. as inability to satisfy focus requirements against a quantificational question.

The second challenging example is in (42), which is due to an anonymous reviewer:

- (42) *Context: In a game of Knuckle, at the end of each round, the person who has more cards than anybody else cashes in and leaves the game. If at the end of a round no player has more cards than anybody else, then all the players stay in the game. Last Saturday night, 5 of my uncles sat down for a long game of Knuckle.*
- a. #I checked in during round three and my uncles were still in the game.
 - b. I checked in during round three and all my uncles were still in the game.

As the reviewer points out, contrastive stress doesn't (visibly) fall on *all* in (42b), so lack of focus with a background quantificational question can't be the problem in (42a); on the other hand, this is again a classic PAI situation since the context rules out **none** (at least some of the uncles must be in the game during round three given the rules of the game).

We will now entertain an alternative route to explaining (41)-(42), which still follows our general dictum that sentences are always answers to a question (Katzir 2024). In the absence of an overt question in both cases, our predictions are highly dependent on what the reconstructed question could be, and our explanation to follow will rely on certain speculations about what the question is, speculations which, without well-understood constraints on question accommodation, are admittedly not easy to entirely justify (but not easy to dispel either). In a nutshell, we will try to explain the infelicitous examples, like WGW do, as a HP+PAI violation—but here the HP will stem from the reconstructed question, rather than triggered directly from the sentence itself. As we will show, due to homogeneity effects, an HP is

expected to arise from the question that a sentence with a definite plural answers if that question is the polar question that contains the sentence and its negation, even if homogeneity effects in general are not due to HP. Our account will then rest on the assumption that in both (41a) and (42a), infelicity comes about due to inability to accommodate questions that are neither quantificational questions nor ones that give rise to HP (and hence to a PAI violation).

Let's first discuss (41). Here the presence of matrix negation could serve as a guide for what the question is: it is plausible to assume that in the absence of other contextual or prosodic cues, sentences with matrix negation and with a default stress pattern signal that the question they answer is the congruent polar question. We then assume that, at least when uttered with default stress, the sentences in (41) are represented with focus on negation, even though this focus doesn't translate to observable stress (presumably because all the other material in the sentence is discourse-new). If that is so, the background questions for the examples in (41) are the following:^{14,15}

- (43) a. Background question for (41a):
 {*John is married to his teammates, John is not married to his teammates*}
- b. Background question for (41b):
 {*John is married to some of his teammates, John is not married to any of his teammates*}

Given Homogeneity of sentences with definite plurals—however it actually arises—the first member of (43a) is equivalent to *John is married to all his teammates* and the second is equivalent to *John is married to none of his teammates*. Questions come with the presupposition that one of their answers is true (see Dayal 1996; Fox 2018b). The presupposition induced by the question in (43a) would then be the **All** ∨ **None** presupposition that either *John is married to all of his teammates* is true or *John is not married to any of his teammates* is. And with this presupposition we would get a PAI violation for the answer in (41a), given that the first disjunct of the presupposition is contextually contradictory. The infelicity of (41a) then comes about as a result of the HP induced by the background question in (43a). The reader can see that this reasoning takes us back almost full circle to WGW's own explanation of the facts in terms of HP—but this time, HP comes from the question rather than directly from the sentence containing the definite plural.¹⁶

Before we continue to our explanation of (42), there is a loophole in our explanation of (41a)

¹⁴ We were inspired to entertain this theoretical direction by Li and Jiang (2026a,b), who propose a similar perspective on the infelicity of certain sentences with plural indefinites (e.g., *Frank doesn't have Roman noses*).

¹⁵ It appears that for some speakers (41a) is not too odd. A possible reason for this is that those speakers can accommodate other background questions than (43a), which would not lead to infelicity, even with default stress. Another possible reason for this could be that speakers can have a metaphoric reading of *married* (as in *he is married to his job*) which would not be expected to lead to infelicity, as being metaphorically married to a group of people is perfectly fine even in a monogamous society. For such speakers, we would expect (i) to be odd (since *born* does not have a comparable metaphoric reading). We thank Brian Buccola for a discussion of this point.

(i) #John wasn't born in these countries.

¹⁶ Note also that the overt yes/no question that corresponds to (43a) is itself infelicitous:

(i) #Is John married to his teammates?

There is a relatively simple explanation for this that is also based on the contradictory nature of the first answer in (43a) given homogeneity. Questions normally come with the presupposition that all their possible answers must be contextually possible (see Fox 2018b, a.o.), so once again, given that the first member in (43a) is equivalent to *John is married to all his teammates*, this presupposition of the question cannot be satisfied.

that we must address. As a reviewer points out, this explanation relies on the assumption that we obligatorily derive a homogeneous (*all*) reading for the positive member of the question in (41a), which raises the question how this can be ensured given that all theories of homogeneity have to explain the existence of so-called ‘non-maximal’ readings, where sentences with definite plurals are interpreted as weaker than their overt *all* counterparts (see Lasersohn 1999; Malamud 2012; Križ 2015, a.o.). A non-maximal reading of the positive member of the question in (43a) would prevent a PAI situation if it is understood non-maximally all-the-way-down, i.e. as plain *some* (‘at least one’). If the positive member is interpreted this way, no problematic presupposition is expected to arise due to the question, and as a result our explanation of the infelicity of (41a) would not go through. We then need to stipulate that the mechanism responsible for deriving non-maximal readings cannot apply here, or at least that deriving the meaning *some* is not possible.¹⁷ This is indeed a potential worry, but we would like to make two points in response. First, there seem to be ill-understood constraints on non-maximality more generally. The emerging consensus is that non-maximal readings for definite plurals are licensed only when the background QUD renders the maximal interpretation irrelevant in some sense; but Križ (2015) pointed out the puzzling fact that sentences with definite plurals cannot have non-maximal *some* readings when they are uttered as direct answers to {*some*, *none*} questions, despite the prediction made by virtually all theories of non-maximality that they should be able to be read that way, and, in fact, Feinmann (2025) notes that the answer here is simply odd:

- (44) a. Did she read any of the books?
 b. ??She read the books.

We do not know why it is impossible to understand the answer here as non-maximal *some* (cf. Feinmann 2025 for an attempt), but we argue that whatever is the reason, it could also account for the impossibility of rescuing (41a) by appealing to non-maximality: in order to avoid infelicity, the positive member of (43a) would have to be interpreted existentially against a {*some*, *none*} question, which, as (44) shows, is impossible.

Second, in the absence of a satisfactory account of the constraints on non-maximality, essentially the same problem affects Guerrini and Wehbe’s account of (41a) as well. For them HP arises due to a presuppositional exhaustivity operator (see more on this in section 7), and non-maximal readings are derived by not having certain alternatives in the domain of this operator (‘pruning’); so that in the absence of these alternatives, HP is not derived. In other words, the HP responsible for the infelicity of (41a) on Guerrini and Wehbe’s account is also only derived under the stipulation that the mechanism responsible for deriving non-maximal readings cannot produce a *some* reading (this point, by the way, applies to their account of (9a) as well). Our theory and Guerrini and Wehbe’s are thus on a par on the non-maximality issue.

Now that we have an account of the infelicity of (41a) in terms of the reconstructed background question (and that we set aside the issue with non-maximality), we are led to the following general prediction: examples of this form should be rendered felicitous if we make adjustments to the context and/or the focus structure that could lead speakers to plausibly accommodate non-problematic questions, i.e. questions that don’t induce HP or ones for which the focus requirement can be met in the answer.

¹⁷ Deriving non-maximal readings that are stronger than *some* for the positive member of the question would lead to infelicity just as if one derived a maximal reading for that sentence, given that it is contextually impossible to marry more than one person.

Let us now attempt to corroborate this prediction.¹⁸

Consider (45), modified from an example by an anonymous reviewer.

- (45) The mayor is friends with some of the people building the highway, but. . .
a. she's not MARRIED to those people (so it may not look good but it's not illegal).

We agree with the reviewer that (45a) is felicitous against this type of context, in contrast to (41a). Note also that (45a) is most naturally read with prominent stress on *married*. The felicity of (45a) is difficult to reconcile with HP+PAI if HP arises from the sentence itself, as WGW assume: Just like in examples (34) and (39a), shifting the focus should not make HP disappear, so a PAI violation is expected to arise in (45a) just like in (41) (though see section 7). On our view, in contrast, one should expect felicity if this example licenses a background question that does not induce an offending HP.¹⁹

Here is what we think may be going on in (45). The issue that arises by this context concerns the legality of the mayor's conduct on the matter of constructing the highway. When this is the issue, the only thing that matters for addressing it is whether the mayor has any legally problematic relations, including but not limited to marriage, with at least one of the people building the highway. Whether she has such problematic relations with one of those people or with more than one is not relevant to that issue, nor whether the problematic relation(s) involves specifically marriage or something else, e.g. siblinghood. Crucially, unlike marriage, one can be problematically related in the relevant sense to more than one person. It is this fact which we think is responsible for escaping a PAI violation here.

One can imagine a number of ways to formally specify a background question that will deliver this, but we can again take the stress pattern of the sentence as our guide. Our hunch is that the focused *married (to)* contrasts with other alternatives that would give other answers to the legality question. Specifically, suppose that *married to* contrasts under negation with something like *unproblematically related to* (or perhaps: *only friends with*), which would give the opposite answer to the legality question.

¹⁸ Somewhat surprisingly from our perspective, it does not look possible to rescue (41a) by introducing an explicit question that is expected to not give rise to HP, together with (observable) polarity focus in the answer, as in (i):

- (i) Q: Is she married to any of her teammates?
A: #(No,) she is NOT married to her teammates.

The reason (i) is surprising in light of our discussion thus far is that it shares relevant properties with example (37), which was judged good (and the only relevant difference between (37) and (i) is in the value of the focused polarity head). Whereas in (37) polarity focus was able to 'rescue' an infelicity with a definite plural against an explicit polar question, here this seems to be more difficult. But whether we predict (i) or not depends on our specific assumptions concerning what makes the question-answer pair in (37) a good one. In our discussion of that example in fn. 11 we entertained two ways to make the answer with the definite plural congruent with the *{all, not all}* background question, one based on the fact that the *not all* alternative was previously mentioned in the discourse, and the other based on either exhaustification or local accommodation below the polarity head. On the first option, we should expect (i) to be just as good as (37), because the *some (=any)* alternative is mentioned in the question. On the second option we should not, since there are only two possibilities as to how to make the sentence congruent with a polar question: We can either derive focus alternatives that are congruent to *{all, none}* (if exhaustification or local accommodation take place above the polarity head), or derive focus alternatives that are congruent to *{all, not all}* (if exhaustification or local accommodation take place below the polarity head); deriving focus alternatives that are congruent to *{some, none}* is impossible, which may explain why (i) is still bad. On the implicature-based account this would require, on top of the assumption that exhaustification cannot be entirely absent from the sentence, as Bar-Lev (2021) assumes, also the assumption that alternatives cannot be 'pruned' here in a way that would derive a *{some, none}* question after all; but that seems to be needed independently, given our discussion of (44) above.

¹⁹ The felicity of (45) also speaks against an alternative account of the infelicity of (41a) that we proposed in a previous version of this paper, according to which it is due to an obligatory computation of contradictory implicatures along the lines of Magri (2009, 2011), which is expected on the implicature account of homogeneity (Magri 2014; Bar-Lev 2021). On that account too, it is not clear why changing the background question should result in felicity.

That is, the interpretation of (45) is something like: she's not married to those people, as opposed to not unproblematically related to them. If that is correct, then the background question triggered by the focus structure, which is moreover consistent with the observed stress pattern, is in (46):

- (46) { *She is not married to those people, She is not unproblematically related to those people* }

Crucially, in contrast to the question in (43a), the question in (46) does not induce a HP that introduces a PAI problem. The resulting presupposition here is merely that either the mayor is not married to (any of) those people, or that she is problematically related to (all of) those people. But, as emphasized above, being problematically related in the relevant sense does not necessarily need to obtain through marriage, and (therefore) does not contextually relate a person to at most one other person. The at-issue content of (45a) is thereby informative even after accommodating the presupposition, and no PAI violation incurs.²⁰

Let us now turn to the other problematic case in (42), repeated in (47).

- (47) *Context: In a game of Knuckle, at the end of each round, the person who has more cards than anybody else cashes in and leaves the game. If at the end of a round no player has more cards than anybody else, then all the players stay in the game. Last Saturday night, 5 of my uncles sat down for a long game of Knuckle.*

- a. #I checked in during round three and my uncles were still in the game.
- b. I checked in during round three and all my uncles were still in the game.

To repeat, the issue is that contrastive stress doesn't (visibly) fall on *all* in (47b), so lack of focus with a background quantificational question doesn't appear to be the problem in (47a).

The challenge (47a) poses for our account is essentially the same as the one (41a) does, and our response is the same: once again, under our account, we must first engage in a "guessing game" about the underlying question, and we expect infelicity if there is no question that can be easily accommodated in a way that the conditions on focus would be met and that would not give rise to HP. What could the reconstructed question be? If it is the polar question *were your uncles still in the game?*, we would expect a question-induced HP which would lead to infelicity, similar to our account of (41) above. If it is the quantificational question *were some or all of your uncles still in the game?*, we would again expect infelicity due to a violation of the conditions on focus. This is not the case in (47b), where one could easily reconstruct the question *were all your uncles still in the game?* from the answer, which would be a good question and the sentence would answer it well.

An obvious question this raises is why it is impossible to accommodate the question *were all your uncles still in the game?* in the case of (47a). We do not have an entirely satisfying answer to this question. We think however that it is reasonable to believe that this question is easier to accommodate for (47b),

²⁰ A crucial ingredient here is that we take negation in (45) not to be within focus (otherwise we would again get the positive contradictory alternative in the question, which may lead to a PAI violation); this may explain why in (i) (due to Brian Buccola), which looks very similar to (45), we once again get infelicity with the definite plural variant in (ia), in contrast with (ib). It is presumably telling that in (ib) negation must be stressed, which indicates that the background question must be a polar one, unlike in (45).

- (i) The mayor's brother is married to one of the people building the highway, but. . .
 - a. #the mayor herself is NOT married to those people. So, it may not look good, but it's not illegal.
 - b. the mayor herself is NOT married to any of those people. So, it may not look good, but it's not illegal.

where *all* appears in the sentence, than in (47a) where it does not. This is not to say that accommodating this question is impossible, but rather our speculation is that it is difficult to accommodate without explicit cues, which is why the sentence looks odd. A possible cue that this question should be accommodated could be provided with polarity focus, which indeed makes (47b) better, similarly to (34):²¹

- (48) In round three I was wondering whether they were all still in the game, so. . .
- a. I checked in and found out that {they, my uncles, all my uncles} were INDEED still in the game.

To conclude, in this section we argued that we can account for cases of infelicity with homogeneity where overt quantificational counterparts do not indicate that only a quantificational question can be accommodated, under the assumption that those cases involve a background polar question that induces HP and, as a result, a violation of PAI. To reiterate, our account of these cases of infelicity with homogeneity is highly speculative and relies on stipulations about what can and cannot be accommodated, which, as we said, are difficult to justify or refute. As such, our account of these effects is not as explanatory as WGW's account based on HP derived from the sentence. This apparent advantage of WGW is however offset by their inability to explain felicitous variants of these examples (as well as those discussed in previous sections). In the next section we will however make a preliminary attempt to make the HP theory compatible with our observations, which, one may hope, could help avoid making the predictions dependent on the guessing game about the background question that we had to engage with in this section.

7 HP with Local Accommodation?

As we mentioned, our account is compatible with various account of homogeneity, except those that assume that HP arises as part of the semantics of sentences containing definite plurals. Our goal in this section is to entertain a possible direction in which even theories that assume HP as part of the semantics can be made compatible with our observations.²²

Recall that our arguments against assuming HP come for instance from examples like (34) and (39a), repeated here as (49) and (50), which show that sentences containing definite plurals can be felicitous even when HP-based theories would predict infelicity due to a violation of PAI.

- (49) A: I know that Mary read some of the books, but I suspect that she didn't read all of them.
B: No, don't worry, she DID read the books.

21 Another option one may entertain, within an implicature view of homogeneity, is that polarity focus allows for the polarity head to take wide scope over the exhaustivity operator that derives homogeneity, in which case the focus value ends up as the {*all*, *not all*} question; and that this is not available without polarity focus because then negation in the polar alternative must take narrow scope relative to exhaustification. This could make sense if the polarity head is higher in the structure than the position of regular negation (see, e.g., Goodhue 2022), and the exhaustivity operator must take scope between these positions. See also fn. 18.

22 A possible reason to save the HP theory has to do with the projection pattern of homogeneity that we mentioned in section 1. On top WGW's argument for HP from infelicity with homogeneity that we focused on, they also provide an argument for it based on projection (see Guerrini and Wehbe 2024: section 6.1). Other authors argued that the projection pattern in fact provides an argument against the HP theory (Spector 2013; Križ 2015; Križ and Spector 2017; but see Fox 2018a for an attempt to reconcile their observations with the HP theory). Since we cannot do justice in this note to the discussion concerning projection, we restrict ourselves to the question whether infelicity with homogeneity can provide an argument either way.

- (50) John really likes to write phonology papers. He also writes semantics papers, but he hates that. Every day, he either works on all of his phonology papers and none of his semantics papers, or all of his semantics papers and some of his phonology papers. He is in a cheerful mood, so apparently. . .
he worked on his PHONOLOGY papers today.

Is there a way to shield the HP theory against these arguments then? One way to avoid PAI violations in these examples on a view that assumes HP could be with a brute-force LF operation responsible for relocating presuppositional information into the assertive component. Such a process is sometimes referred to as Local Accommodation (LA) (Heim 1983, Beaver and Krahmer 2001). If, for some reason, LA is possible in (49) and (50), this could explain why PAI isn't violated in these examples despite there being HP: Once LA applied, the sentence would no longer carry HP and therefore there would be no PAI violation. Of course, one would have to say something about the distribution of LA, since if LA was always possible we would never expect PAI effects to arise (nor would we ever be able to detect presuppositions as such); the challenge would then be to explain what makes LA possible in cases like (49) and (50).

One may look for clues in the non-default stress pattern in these examples. Indeed, Bassi et al. (2021) observed that LA when computed under negation must be accompanied by a particular stress pattern. Furthermore, as we pointed out in footnote 5, certain stress patterns are able to rescue from PAI violations with (run-of-the-mill) presuppositions. These observations could be explained if stress is a condition on the application of LA. (51)-(53) illustrate this further with the triggers *stop*, singular *the* (uniqueness presupposition), and *again*, respectively.²³

- (51) I knew that Mary doesn't smoke now, but today I learned that...
a. she {STOPPED, #stopped} smoking. (She used to be a heavy smoker!)
- (52) I knew that Mary is planning to speak in the conference about a book she wrote. But today I learned that...
a. she will be speaking about {THE, #the} book she wrote. (She has just one!)
- (53) I already knew that Mary is on sabbatical this year. But today I realized that...
a. she's on sabbatical {AGAIN, #again}. (She took a sabbatical before, she's so lucky!)

Note that stress on anything else in (51)-(53) leads to infelicity. The generalization from these data seems to be that LA is possible only if main stress falls on the relevant presupposition trigger, and this is in line with a similar claim made in Bassi et al. (2021). If that is so, then the inability of the trigger of HP in definite plurals to bear focal stress, the factor which we identified as relevant for explaining infelicity with homogeneity (sections 4 and 5.2), could now also be thought of as the reason why LA cannot save infelicity with homogeneity in most cases.

But even so, we are still short of explaining what's going on in the felicitous examples (49) and (50), because these examples do not pattern accordingly: in (49) there is polarity focus, and in (50) stress falls on some nominal within the definite plural; presumably, in neither case could the stressed element be

²³ See Doron and Wehbe (2024) for some relevant discussion of LA and PAI (though they do not discuss the ability of stress to rescue PAI violations).

considered the HP trigger. This however is not to say that the direction we just sketched, in terms of a stress condition on LA, is hopeless against the challenge posed by (49) and (50). Below is a sketch of a possible speculative direction worthy of pursuit if one continues this line of thought.

This direction is based on combining the implicature account of homogeneity with the ‘presuppositional implicatures’ account by Bassi et al. (2021), which is precisely the account Guerrini and Wehbe (2024) assume. On the implicature account in Magri (2014); Bar-Lev (2021), an exhaustivity operator $\mathcal{E}xh$ is responsible for implicature computation, and the homogeneity inferences that $\mathcal{E}xh$ computes are at-issue content. Bassi et al. (2021) however argue that implicatures are non-at-issue content, and accordingly propose that implicature computation is done by $\mathcal{P}ex$, an exhaustivity operator that presupposes the inferences that $\mathcal{E}xh$ is usually taken to assert. Once Bassi et al.’s $\mathcal{P}ex$ theory is combined with the implicature account of homogeneity, as Guerrini and Wehbe assume, homogeneity is predicted as a presupposition.

On Guerrini and Wehbe’s account, the trigger of HP is $\mathcal{P}ex$, which associates with a covert pluralization operator. One may ask then in general what stress pattern allows for LA when the presupposition trigger is $\mathcal{P}ex$: It can’t possibly be stress on $\mathcal{P}ex$ itself, which is covert. The generalization seems to be, based on Bassi et al.’s own observation, that LA’s stress condition is satisfied if stress falls on a focus associate of $\mathcal{P}ex$. One can then imagine that, when there are several alternative-generating items that $\mathcal{P}ex$ associates with, stress on one of them should be enough to allow for LA. If that is on the right track, one could explain the felicity of (50) by assuming that $\mathcal{P}ex$ associates with both *phonology* and the covert pluralization operator, so stress on *phonology* suffices to license LA. While this would work for (50), the case of (49) is more difficult: explaining its felicity along the same lines would presumably require assuming that $\mathcal{P}ex$ associates with both the covert pluralization operator and the polarity head; whether this could be made to work within a reasonable account of polarity focus is a question we will leave to another occasion.²⁴

8 Concluding remarks

To summarize, we proposed that infelicity with homogeneity is not due to an HP encoded in the semantics of sentences containing definite plurals as WGW argued, but rather due to conditions on focus. We argued that a focus-based explanation has advantages over the HP+PAI view: (i) it can explain infelicity in some cases where HP+PAI is silent ((21b-i) and (22a)), and (ii) it can explain felicity in some cases where HP+PAI predicts infelicity ((49) and (50)). We further argued that the only cases that require assuming that there is HP after all (Schwarzschild’s example in (41a), as well as (42a)) can be explained by attributing HP to the background question (QUD) rather than the sentence itself.

Before concluding, we would like to mention another potential argument for the focus-based perspective, coming from other domains where homogeneity effects have been observed. Not all of them behave uniformly with respect to infelicity with homogeneity, as WGW point out. The sentences in (54) (based on example (44) in Guerrini and Wehbe 2024) replicate WGW’s observation with definite plurals in other domains: Habituals ((54a); see Agha 2023), generics ((54b); see Löbner 2000; Križ 2015; Agha 2023),

²⁴ For a view where the polarity head is a focus associate of an exhaustivity operator see Bassel (2023). For her purposes this operator must take scope above an assertion operator, whereas the exhaustivity operator that is responsible for homogeneity presumably takes scope below an assertion operator; this makes it difficult to explain the felicity of (49) as stemming from a single exhaustivity operator that associates with both the polarity head and the operator that triggers homogeneity.

episodic universal bare plurals ((54c); see [Condoravdi 2016](#); [Chierchia 2022](#)), predicate homogeneity ((54d); see [Križ 2015](#); [Paillé 2024, 2025](#)), and conditionals ((54e); see [von Stechow 1997](#); [Herburger 2015](#); [Križ 2015](#); [Bassi and Bar-Lev 2016](#); [Agha 2023](#)).

- (54) a. I knew that Mary sometimes goes to the gym on Tuesdays, but today I learned that. . .
 (i) #Mary goes to the gym on Tuesdays.
 b. I knew that wolves sometimes hunt in packs. But today I learned that. . .
 (i) #wolves hunt in packs.
 c. I knew that some birds are migrating right now. But today I learned that. . .
 (i) #birds are migrating right now.
 d. I knew that the wall was partly red. But today I learned something interesting. . .
 (i) #the wall is red.
 e. I knew that if you hit the brakes the car sometimes stops, but today I learned that. . .
 (i) #if you hit the brakes, the car stops.

Parallel examples in other domains that show what looks like homogeneity effects, however, behave differently. Specifically, with free choice disjunction ((55a); see particularly [Goldstein 2018](#); [Bar-Lev 2018, 2021](#) for the connection between free choice and homogeneity) and with neg-raising ((55b-d); see [Gajewski 2005](#); [Staniszewski 2019](#); [Jeretič 2021](#); [Agha and Jeretič 2022](#)), there is no infelicity in parallel contexts, as long as the sentence is uttered with a particular stress pattern, as (55) demonstrates ((55a) and (55d) are based on [Guerrini and Wehbe 2024](#), exx. (42) and (47)):

- (55) a. I knew that Mary can eat the cake and I wasn't sure about the candy. But today I learned that. . .
 (i) she can eat the cake {OR, #or} the candy.
 b. I knew that Mary is fine with meeting over zoom, but today I learned that. . .
 (i) she {WANTS, #wants} to meet over zoom.
 c. I knew that Mary thinks it's possible that the earth is flat, but today I learned that. . .
 (i) she (actually) {BELIEVES, #believes} the earth is flat.
 d. I knew that Mary is allowed to read this book, but today I learned that. . .
 (i) she {SHOULD, #should} read this book.

If it is desirable to maintain a unified treatment of homogeneity phenomena across domains, one may take the contrast between (54) and (55) as support for the view we proposed here (as [WGW](#) mention), given that the distinction between free choice and neg-raising on the one hand and all the other domains on the other hand tracks the difference between having an overt connective or quantificational element (*or* in the case of free choice, *want/believe/should* in the case of neg-raising) and not having any such thing (all other domains), and that stress on these overt items is required. Whether this is indeed a strong argument in favor of our approach is not entirely clear to us for two reasons: First, the verdict is still out on whether homogeneity phenomena should be treated uniformly. Second, as we discussed in section 7, PAI violations can be avoided with stress on presupposition triggers, which potentially provides an alternative explanation for the felicity of the examples in (55) on accounts that assume HP.²⁵

²⁵ This would presumably require assuming that *or*, *want*, *believe*, and *should* are either the triggers of HP in the examples in (55), or are focus associates of a presuppositional exhaustivity operator that derives HP (based on our discussion in section 7).

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This may provide a way to tease apart theories of homogeneity effects in these domains: For instance, on Goldstein’s (2018) account of free choice disjunction, the trigger of HP is the modal, which makes it surprising that stress falls on *or* in (55a-i). In contrast, this is expected if disjunction is the focus associate of a presuppositional exhaustivity operator that derives HP, as in Del Pinal et al. (2024).

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A F-placement vs. F-realization

As discussed in section 5.2, when the question is a quantificational *some/all* question, an explanation of infelicity with homogeneity that is based on failure to satisfy requirements from focus is compatible with all theories of homogeneity. The question remains what kind of failure it is exactly—one due to F-placement or one due to F-realization. For convenience the basic explanandum is repeated in (56) and the two principles are repeated in (57)-(58):

- (56) a. Did Ann read none of the books, some of the books, or all of them?

b. #Ann read the books.

(57) **Focus Placement:**

- a. **Question-answer congruence:** The focus structure of a sentence S must be congruent to an explicitly-provided or implicitly-accommodated question Q that S answers, such that $Q \subseteq \llbracket S \rrbracket_f$.
- b. **Focus minimization:** F-mark as little as is necessary to satisfy question-answer congruence.

(58) **Focus Realization:**

The highest stress in a sentence (more generally, in a focus domain) lies on an F-marked phrase in the sentence. (adapted from Büring 2016:164)

As we mentioned in section 5.2, on theories where nothing covert in the structure contributes quantification (e.g., Križ 2015; Križ and Spector 2017), the failure in (56) would have to be due to F-placement. In contrast, on theories of homogeneity where it is triggered by covert operators (Gajewski 2005; Bar-Lev 2021; Guerrini and Wehbe 2024), it's also possible to attribute the infelicity to F-realization. In this appendix we discuss whether these accounts predict the problem to arise at the level of F-realization or F-placement. As it turns out, this depends on the specific assumptions regarding the semantics and focus alternatives of the operator in question.

Let us proceed on the supposition that homogeneity is triggered by a covert operator. If the quantificational force of the relevant operator is universal (a distributivity operator), as in Gajewski (2005), then the satisfiability of F-placement depends on assumptions concerning its focus alternatives: If one assumes that it has an existential focus alternative, F-placement would be satisfied but F-realization would not, and if it doesn't then the failure will arise already at the level of F-placement. The situation is more complicated on the implicature account of Homogeneity in Bar-Lev (2021) and Guerrini and Wehbe (2024), which crucially assumes that the relevant operator is existential and does not have a universal alternative. Specifically, one may expect a failure of F-placement on this account for the following reasons. *She read the books* has the following focus alternatives before implicature computation, assuming there are two books, b_1 and b_2 . (We assume here that the domain variable is within focus, but having other constituents within focus would not change the picture, given the assumption that more complex alternatives are not derivable, as in Fox and Katzir 2011, which is crucial for the implicature account of homogeneity.)

(59) the books $\exists\text{-PL}_{[D]_F} \lambda x$ She read x

- | | |
|--|------------------------------------|
| a. the books $\exists\text{-PL}_{\{b_1, b_2\}} \lambda x$ She read x | She read $b_1 \vee$ She read b_2 |
| b. the books $\exists\text{-PL}_{\{b_1\}} \lambda x$ She read x | She read b_1 |
| c. the books $\exists\text{-PL}_{\{b_2\}} \lambda x$ She read x | She read b_2 |

Note that this set of focus alternatives is not congruent to the quantificational *some/all* question, since no alternative in (59) is equivalent to the *all* alternative in the question. This changes once we consider the result after implicature computation, since the meaning after implicature computation, in (60), is equivalent to the *all* alternative (on Guerrini and Wehbe's account, of course, $\mathcal{E}xh$ here should be replaced with $\mathcal{P}ex$):

- (60) $\mathcal{E}xh(\text{the books } \exists\text{-PL}_{[D]_F} \lambda x \text{ She read } x)$
- | | | |
|----|---|---|
| a. | $\mathcal{E}xh(\text{the books } \exists\text{-PL}_{\{b_1, b_2\}} \lambda x \text{ She read } x)$ | She read $b_1 \wedge$ She read b_2 |
| b. | $\mathcal{E}xh(\text{the books } \exists\text{-PL}_{\{b_1\}} \lambda x \text{ She read } x)$ | She read $b_1 \wedge \neg$ She read b_2 |
| c. | $\mathcal{E}xh(\text{the books } \exists\text{-PL}_{\{b_2\}} \lambda x \text{ She read } x)$ | She read $b_2 \wedge \neg$ She read b_1 |

However, this set of alternatives is still not congruent to the quantificational *some/all* question, since now no member of the set of alternatives corresponds to the *some* alternative in the question. In other words, congruence is satisfied neither before nor after exhaustification if the set of focus alternatives is as in (59) before exhaustification and as in (60) after.

In light of this, one may think that on the implicature account the relevant failure here must be due to F-placement. However, even within the implicature account there are ways to make F-placement satisfiable after all. One way to do that would be to assume that the set of focus alternatives of (60) contains alternatives where $\mathcal{E}xh$ is deleted; in that case, its set of focus alternatives would be the union of (59) and (60) and congruence would be satisfied. Another way to do that would be to relax the requirement on congruence so that every member of the question must be equivalent to a *disjunction* of alternatives in the focus value of the sentence; in that case, congruence would be satisfied because the disjunction of alternatives in (60) is equivalent to the *some* alternative in the question.

Do we have reasons to assume that the failure here is due to F-placement or due to F-realization? We are not sure. As a reviewer notes (and as we mentioned in section 5.2), assuming that the failure here is due to F-realization requires assuming that F-realization is an inviolable interface constraint, which one may want to avoid (though this is what is assumed also in Büring 2016: p. 184). We should however note that a view where the culprit is F-placement, while promising, would face difficulty explaining cases where the presence or existence of overt material seems to make a difference; recall our discussion of the contrast between (54) and (55), which is straightforwardly explained if the failure in (54) is due to F-realization, and it is avoided when there is overt material that can bear stress as in (55); we do not know how to make sense of this contrast if the problem is due to F-placement, at least without stipulating that stress in these examples somehow avoids the issue with F-placement (e.g., by assuming that stress on the associate of $\mathcal{E}xh$ somehow allows for deleting it and thus satisfying congruence).